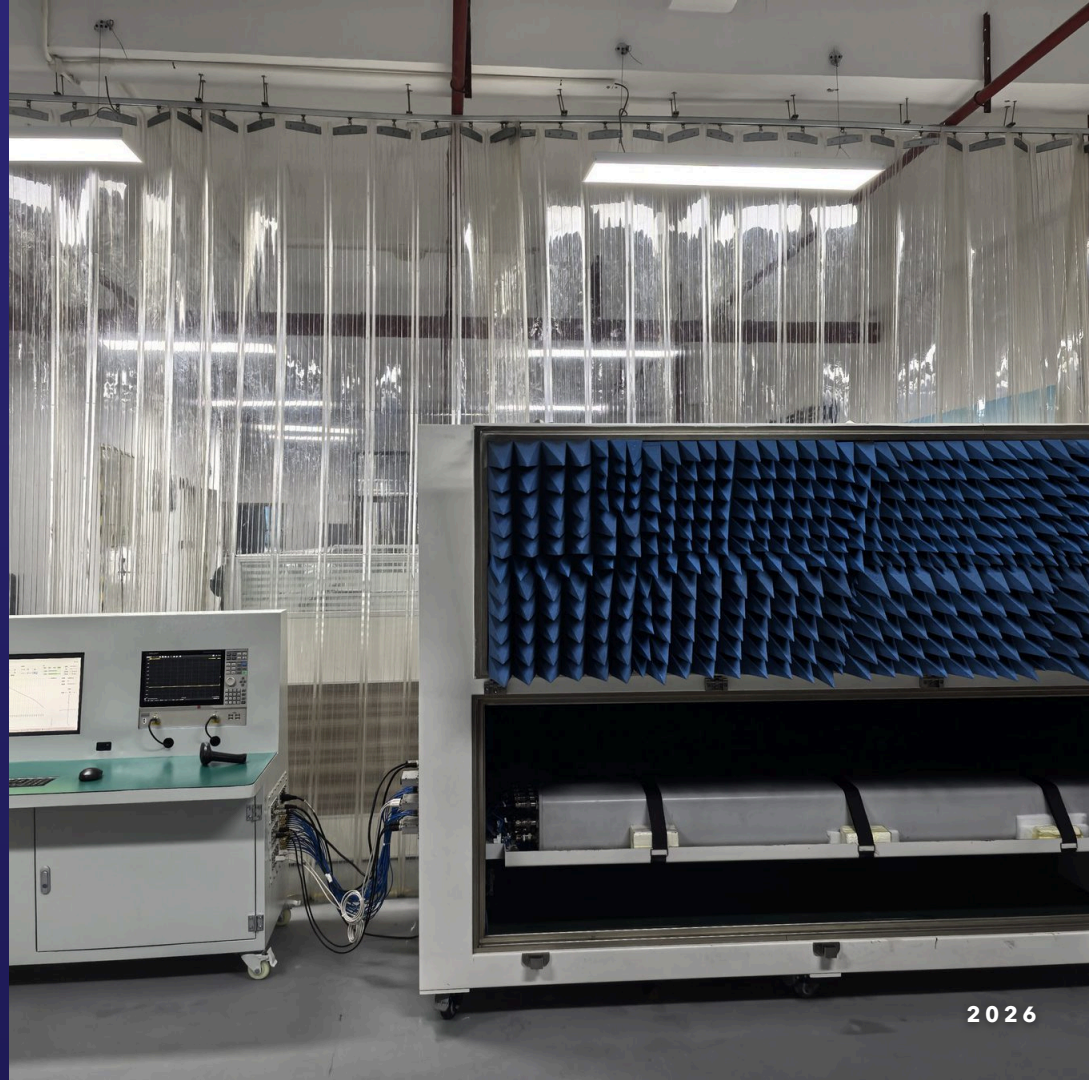
Company Profile

Fast, reliable and cost-effective testing for base station antennas, backed by almost three decades of telecom network service.



2026

Dongguan Voyager Telecom Equipment CO., LTD.
Xi'an Voyager Telecom Equipment CO., LTD. (Joint Stock)

AT A GLANCE

Who we are

VOYAGER TELECOM

Voyager is a Chinese telecom test and service company with almost three decades of experience. Xi'an Voyager Telecom Equipment CO., LTD. (Joint Stock), founded in 1997 and recognised as a National High-Tech Enterprise, built its business on in-house test systems and repair services that keep telecom networks running, for operators such as China Mobile and China Telecom and equipment makers such as Huawei and ZTE.

In 2021 Xi'an Voyager and senior professors from the Institute of Antennas and Electromagnetic Scattering at Xidian University established Dongguan Voyager Telecom Equipment CO., LTD. It combines nearly 30 years of test-system development with deep antenna theory and antenna test engineering experience, and develops cost-effective intelligent antenna test solutions under the **ANTestify** brand, helping manufacturers reduce antenna testing costs and operators reduce antenna O&M costs.

CORE BUSINESS

- **Antenna test equipment:** R&D, manufacturing and sales
- **Third-party antenna testing services**
- **Antenna repair and refurbishment services**

1997
Founded in Xi'an, China

29
Years of telecom testing and network services

6
Product series of telecom equipment test systems

4 in 1
Radiation, circuit, PIM and RET tests in one station

≈20 min
Complete automatic test of a 4+4 antenna

10–50×
Higher test efficiency than conventional methods

2,000+
Antennas refurbished for China Mobile Hainan

PCT
International patent application filed; 9 registered software copyrights

OUR JOURNEY

From network maintenance to antenna testing

Each stage of our history builds on the test and service capability of the one before.

1997
Xi'an Voyager is founded, one of the earliest telecom equipment repair companies in China.

2000
The founder's entrepreneurial story is featured by China Central Television (CCTV).

2000s
Six series of in-house test systems: switching, PHS base stations, telecom power, digital microwave, broadband access and GSM/CDMA base stations.

2010s
GSM base station simulation and telecom power test platforms appraised by the O&M Committee of the China Association of Communication Enterprises.

2012-15
A Voyager service team based in Algiers, Algeria, provides telecom equipment repair as an independent repair company.

2021
Dongguan Voyager is founded with antenna experts from Xidian University.

2024
Antenna testing, repair and refurbishment programme starts for China Mobile Hainan.

2025
Strategic cooperation agreement with CTTL. Repair contract for repeaters and antennas with China Telecom Xi'an (2025–2027).

2026
CTTL measurement audit of gain measurement capability. International market launch.

Network services heritage

Nearly three decades of testing, repairing and maintaining switching, transmission, power and base station equipment for operators and equipment makers.

Test-system engineering

Every repair line is backed by test systems we design ourselves, the same engineering approach behind the TTC3000 antenna test family.

Antenna specialisation

Since 2021, a dedicated antenna company combining Voyager's system engineering with Xidian University's antenna measurement research.

WHAT WE STAND FOR

Mission, vision and values

VOYAGER TELECOM

MISSION

Make complete antenna testing fast and affordable enough to apply to every antenna: at the factory, at the warehouse, before installation and after repair.

VISION

To be the testing partner that antenna manufacturers and network operators worldwide rely on, so networks perform better at lower cost and with less waste.

RELIABILITY

Built to keep networks running

Our roots are in network maintenance. We design equipment for daily, repeatable use.

ENGINEERING

Measured, not assumed

Methods follow national and industry antenna standards and are checked against independent laboratories.

PARTNERSHIP

Long-term relationships

We work with customers from first trial to daily operation, with training and ongoing support.

SUSTAINABILITY

Repair before replace

Refurbishment turns idle antennas back into network assets and cuts electronic waste.

THE INDUSTRY CHALLENGE

Antenna testing has not kept pace with the network

VOYAGER TELECOM

- **Production at scale.** With 5G in deep global deployment, antenna lines now complete an antenna every few minutes. Laboratory-grade testing cannot keep up, so full inspection is rarely possible.
- **Networks at scale.** A mature network holds hundreds of thousands of sites. Antennas outdoors degrade over time, and a VSWR check alone cannot locate radiation faults.
- **Conventional methods are slow and fixed.** Far-field ranges and multi-probe spherical near-field systems need dedicated buildings, high investment and days of measurement for a large multi-port antenna.

Under the guidance of the Antenna and RF Branch of the China Institute of Communications, Voyager developed rapid test technology and equipment for passive base station antennas that is efficient, reliable, easy to operate and low in cost: a bridge between intelligent antenna manufacturing and intelligent network O&M.

For manufacturers it enables "last mile" full or high-frequency sampling inspection, improving consistency and brand competitiveness. For operators it supports prevention beforehand, rapid diagnosis during operation and accurate evaluation afterwards, reducing network downtime and optimising CAPEX and OPEX.

MANUFACTURERS

Factory outgoing inspection

Connects directly to the production line for full or high-frequency sampling inspection before shipment.

OPERATORS

Arrival inspection

Checks antenna quality at operator warehouses on delivery, protecting procurement quality.

OPERATORS

Pre-installation testing

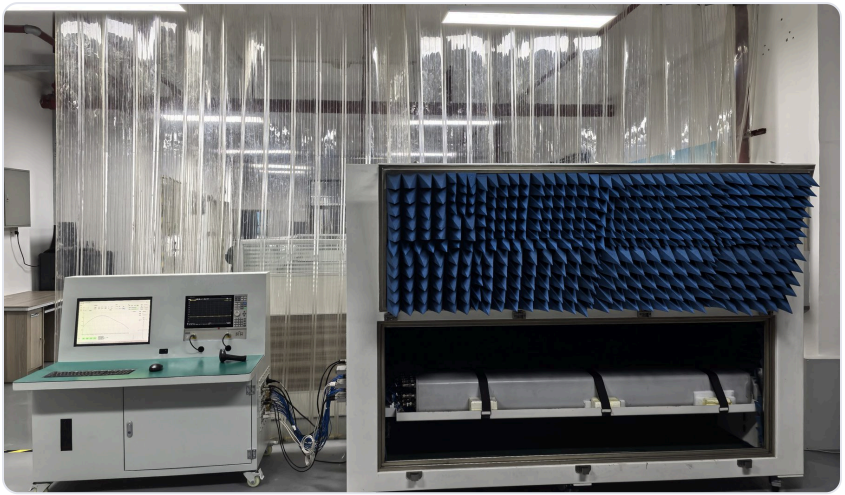
The vehicle-mounted platform tests antennas on site before tower installation, avoiding a second climb.

SERVICE COMPANIES

Repair and refurbishment

Automated testing, fault analysis and assisted fault location so repaired antennas can return to the network.

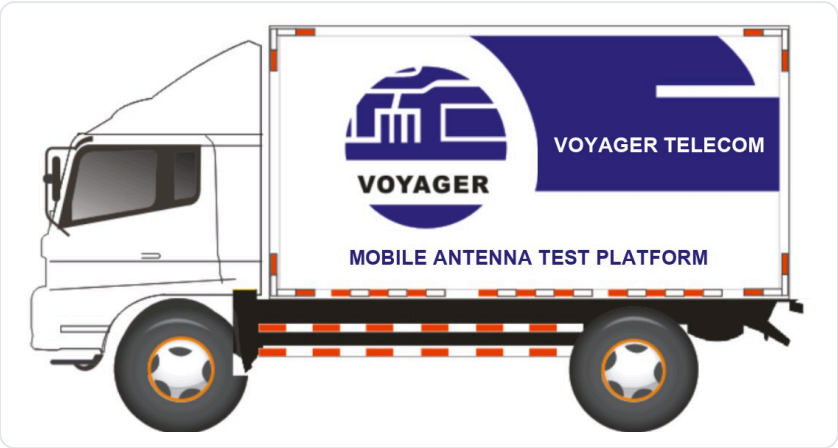
TTC3000 Base Station Antenna Integrated Testing System



Model	Test functions	Frequency range
TTC3000-B4G / B8G	RET interface + circuit parameters	600 MHz–4 GHz / 8 GHz
TTC3000-H4G / H8G	RET + circuit + PIM	600 MHz–4 GHz / 8 GHz
TTC3000-R4G / R8G	RET + circuit + PIM + radiation	600 MHz–4 GHz / 8 GHz

- **All-in-one and compact.** Radiation, circuit, PIM and RET interface testing integrated in one system with a footprint under 10 m². No dedicated indoor anechoic room is required.
- **Production-line speed.** All electrical parameters of an antenna are tested at one station. A complete automatic test of a 4+4 antenna takes about 20 minutes, and of a 4+4+8+8 antenna about 40 minutes: 10–50 times the efficiency of conventional methods.
- **Automated end to end.** Scan the product barcode and the system identifies the antenna model, loads the test script, switches ports, sets electrical downtilt and produces the test report.
- **Easy operation.** No alignment to the antenna phase centre and no manual port changes, greatly reducing labour.
- **Low cost to build and run.** Typically 10–30% of the price of conventional systems. Works at room temperature (25 ± 5 °C), and users can maintain, service and calibrate the system themselves.
- **Full traceability.** Test reports are uploaded automatically, keeping data for the whole product life cycle.

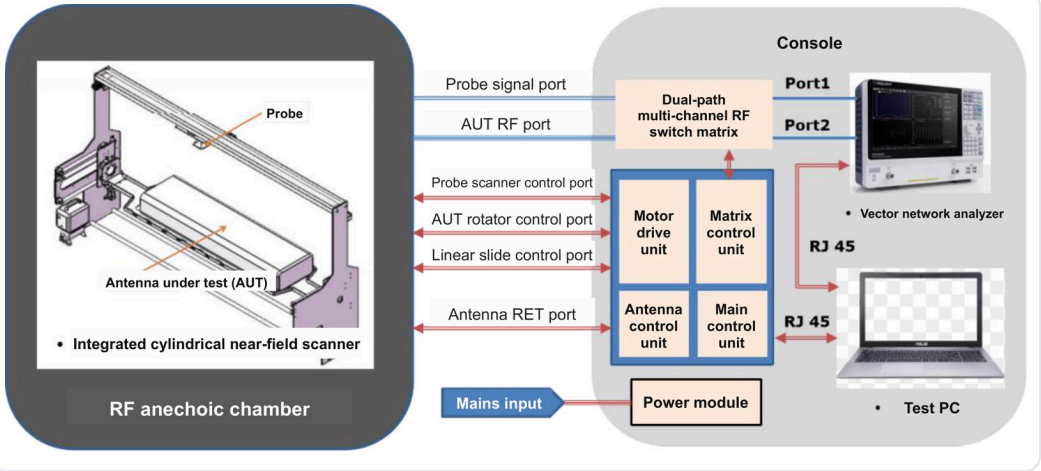
Vehicle-Mounted Base Station Antenna Testing Platform



Model	Application	Vehicle
TTC3000-C4G / C8G	Urban, external power	Light truck or van
TTC3000-U4G / U8G	Suburban, on-board power	Medium truck or bus
TTC3000-W4G / W8G	Field, on-board power and air conditioning	Heavy truck or coach

- **Complete test coverage.** Tests circuit parameters such as VSWR, isolation, PIM, coupling and electrical downtilt, and quickly measures radiation parameters such as gain and beamwidth.
- **Mobile and flexible.** The vehicle-mounted RF anechoic chamber is built with refrigerated-truck body techniques for thermal insulation and sun protection. It works in the field and suits pre-installation testing and rapid network restoration after disasters.
- **Five-stage vibration isolation.** Air suspension on the chassis plus vibration-damped mounting of the scanner, console and instruments allows travel on rural roads.
- **Self-contained power and air conditioning.** The field model operates independently for 10 hours or more.
- **Independently assessed.** The platform's gain measurement capability was assessed in a CTTL measurement audit.

Fully automatic electrical parameter test system



Voyager's system combines a movable RF anechoic chamber, an integrated cylindrical near-field scanner, a dual-path multi-channel RF switch matrix and a vector network analyzer under one control console. Near-field data are transformed to far-field patterns in software, so a large multi-port antenna can be characterised without a far-field range.

Chamber, scanner, switch matrix and software are developed in-house. The core hardware is patent-pending, with patent applications in China and a PCT international application, and the software is protected by 9 registered software copyrights.

Movable RF anechoic chamber

Frequency range	**100 MHz–8 GHz**
Shielding	**≥ 90 dB**
Reflectivity	**≤ –30 dB**

Probe scanner

Travel	**0–3000 mm**
Positioning accuracy	**±0.05 mm**
Speed	**≥ 100 mm/s**

AUT rotator

Rotation range	**–180° to +180°**
Positioning accuracy	**±0.05°**
Speed	**≥ 5°/s**

Near-field to far-field

Agreement with theory	**≥ 97%**
PIM analyzers	**Up to 8**
24-port antenna, full scan	**< 10 hours**

Complete parameters, automatic workflow

VOYAGER TELECOM

Circuit parameters

VSWR, isolation, calibration-port coupling factor, amplitude and phase deviation.

Passive intermodulation

PIM testing with up to 8 external PIM analyzers connected.

Radiation parameters

Radiation pattern, gain, beamwidth, front-to-back ratio, cross-polar discrimination (CPD), upper sidelobe suppression, null fill, elevation downtilt deviation, sector power ratio, azimuth beam port-to-port tracking and azimuth beam squint.

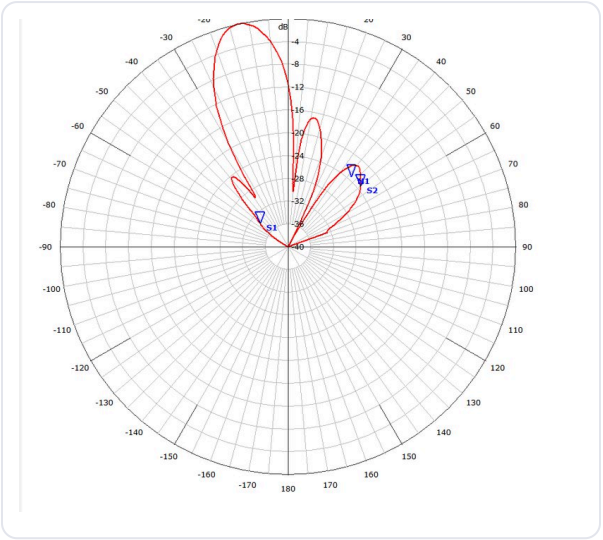
RET interface

RET control unit search and matching, electrical downtilt setting.

Reports and data

Word and Excel reports with 20+ pattern parameters per port and frequency, uploaded automatically.

Scan barcode → **Identify model** → **Load test script** → **RET and circuit tests** → **Radiation tests** → **Report and upload** → **Fault classification**



Supported antenna types include multi-band independent-RET smart antennas (such as 2+2+8+8 and 4+4+8+8 configurations), conventional and high-gain smart antennas, 8-channel RET antennas and multi-band FDD/NB-IoT RET antennas.

INDEPENDENT VALIDATION

CTTL measurement audit of gain measurement capability



China Telecommunication Technology Labs (CTTL), part of the China Academy of Information and Communications Technology (CAICT), conducted a measurement audit of the gain measurement capability of Voyager's Vehicle-Mounted Testing Platform.

Method

Stable low-band and high-band base station antennas were measured, and the results were compared with the robust average of the data from China's major testing laboratories.

Result

The gain measured by Voyager's platform is consistent with the results of the mainstream laboratories. Detailed test data are given in the Result Report.

Strategic cooperation

Voyager and CTTL signed a strategic cooperation agreement in November 2025, and CTTL requested a TTC3000 system for reliability verification.

Technical basis

GB/T 9410 general specification for mobile communication antennas, YD/T 2867 and YD/T 2868 passive base station antenna requirements and measurement methods.

CTTL was established in 1981 and provides testing, certification, metrology and consulting services to government bodies, operators and manufacturers in the ICT sector.

One partner from equipment to service

EQUIPMENT

Antenna test systems

TTC3000 integrated testing systems and vehicle-mounted platforms, with installation, calibration and operator training.

TESTING

Third-party antenna testing

Independent testing of base station antennas for operators and manufacturers, including arrival and pre-installation inspection.

CIRCULAR ECONOMY

Antenna repair and refurbishment

End-to-end testing, repair and refurbishment of decommissioned and faulty antennas, restored to the standards required for network re-entry.

NETWORK SERVICES

Telecom equipment repair and maintenance

Testing, repair and maintenance of repeaters, base stations, telecom power and transmission equipment, supported by our own test platforms.

SOFTWARE AND DATA

Test data management

Test scripts, reports and life-cycle data management, with fault classification and assisted fault location for repair lines.

SUPPORT

Training and technical support

Operator training, test-method training and long-term technical support for every system we deliver.

CASE STUDY · CHINA MOBILE HAINAN

Repair instead of replace

VOYAGER TELECOM

CHALLENGE

Idle and faulty antennas

Decommissioned and faulty antennas were piling up. The traditional practice of replacing rather than repairing meant high procurement costs and growing electronic waste.

APPROACH

Testing-led refurbishment

China Mobile Hainan commissioned Voyager to provide end-to-end professional testing, repair and refurbishment with our own equipment. A standardised repair process fully restores every technical specification to the standard for network re-entry: repair instead of replace.

01 · TWO-YEAR CUMULATIVE

Idle assets revitalised

Decommissioned and faulty equipment regains its value, raising fixed-asset utilisation and reducing idle resources and waste.

02 · TWO-YEAR CUMULATIVE

Procurement costs reduced

Refurbishment costs only 30–50% of buying new, cutting hardware investment and improving cash flow.

03 · TWO-YEAR CUMULATIVE

Green development practised

Less electronic waste and pollution and lower manufacturing emissions, supporting corporate social responsibility.

2,000+
Antennas refurbished since November 2024

30–50%
Refurbishment cost relative to buying new

100%
Of technical specifications restored for network re-entry

End-to-end
Testing, repair and refurbishment by one team

CUSTOMERS AND PARTNERS

Who we work with

VOYAGER TELECOM

China Mobile

OPERATOR

Network equipment services and, in Hainan, antenna testing, repair and refurbishment since 2024.

China Telecom

SERVICE CONTRACTOR

Repair of wireless repeaters and wide-beam antennas for China Telecom Xi'an, 2025–2027.

Huawei

LONG-TERM SERVICES

Testing, repair and maintenance services for telecom equipment, including repeater equipment.

ZTE

LONG-TERM SERVICES

Testing, repair and maintenance services for telecom network equipment.

CTTL · CAICT

STRATEGIC COOPERATION

Measurement audit of gain measurement capability and strategic cooperation agreement, 2025.

Xidian University

ACADEMIC PARTNER

Senior professors of the Institute of Antennas and Electromagnetic Scattering co-founded Dongguan Voyager.

China Institute of Communications

TECHNICAL GUIDANCE

The Antenna and RF Branch guided the development of our rapid passive antenna test technology.

Your company?

REGIONAL PARTNERS WANTED

We are looking for distributors, testing-service partners and refurbishment partners in new markets.

Founder and Chairman



Zhu Jianping

Founder & Chairman, Xi'an Voyager Telecom Equipment CO., LTD. (Joint Stock)

General Manager, Dongguan Voyager Telecom Equipment CO., LTD.

- Master's degree in Communication Engineering from Huazhong University of Science and Technology; Senior Engineer.
- Began his career at the 10th Research Institute under China's Ministry of Industry and Information Technology.
- Founded Xi'an Voyager in 1997 and has led it for almost three decades, building one of China's earliest telecom equipment repair and test-system businesses.
- Has long led the development of telecom equipment test systems and is regarded as a leading figure in China's telecom operation and maintenance sector. His entrepreneurial story was featured by China Central Television (CCTV) in 2000.
- Leads Voyager's move into base station antenna testing and international markets.

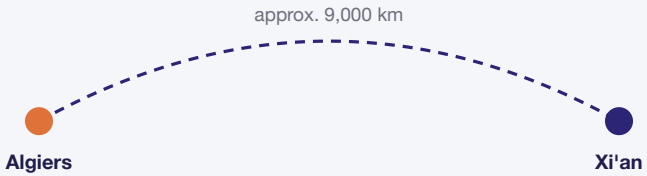
1997	29 yrs	2000	2021
Founded Xi'an Voyager	Leading the company	Featured by CCTV	Launched Dongguan Voyager

On the ground in Algiers, Algeria

VOYAGER TELECOM

2012–2015

ALGIERS · ALGERIA



From 2012 to 2015 Voyager stationed a service team in Algiers, the capital of Algeria, to repair telecom network equipment in service. Voyager operated as an independent repair company, alongside the project teams of Chinese equipment vendors, including Huawei.

INDEPENDENT

Our own repair operation

Voyager worked as a separate repair company, with its own engineers, test platforms and repair processes.

ALONGSIDE MAJOR VENDORS

Supporting networks in service

The team worked alongside vendor project teams, including Huawei's, supporting telecom network equipment in service.

ON SITE

Based in the capital

Repair services were delivered by a team stationed in Algiers, close to the networks and customers they served.

READY FOR NEW MARKETS

Experience abroad

Running an independent repair operation far from home shaped how we support customers and partners in overseas markets today.

Partnering with Voyager

Distribution and agency

Regional partners represent the TTC3000 family to operators, tower companies, managed-service providers and antenna manufacturers.

Local testing services

Partners operate Voyager systems to offer arrival, pre-installation and in-service antenna testing in their market.

Refurbishment programmes

Joint antenna repair and refurbishment programmes based on the model proven with China Mobile Hainan.

Factory quality control

Production-line testing for antenna manufacturers serving the region, wherever they produce.

Training and technology transfer

Operator training, test-method training and long-term technical support to build local testing skills.

WHAT PARTNERS RECEIVE

Everything needed to test locally

- **Equipment and test scripts** configured for the antennas in your market.
- **Installation and calibration** on site, ready for daily testing.
- **Training** for operators and engineers on test methods and maintenance.
- **A proven service model** for testing and refurbishment, based on our work with China Mobile Hainan.
- **Long-term technical support** from the team that designs the system.

Why partners choose Voyager

01

Almost three decades in telecom

Founded in 1997, with long-term service relationships with China Mobile, China Telecom, Huawei and ZTE.

02

Independently assessed

Gain measurement capability assessed in a CTTL measurement audit against China's major testing laboratories.

03

Speed and cost

A 4+4 antenna fully tested in about 20 minutes, 10–50 times the efficiency, at typically 10–30% of the price of conventional systems.

04

Factory, warehouse or field

The same test system in a compact chamber or on a truck, with on-board power and air conditioning for field work.

05

Proven in operation

More than 2,000 antennas refurbished for China Mobile Hainan, at 30–50% of the cost of new antennas.

06

In-house technology

Chamber, scanner, switch matrix and software developed in-house with Xidian University antenna experts; patent-pending, including a PCT application.

CONTACT

Let's work together

We welcome enquiries from operators, tower companies, antenna manufacturers, repair companies and regional partners.

EMAIL **zsy@voyager8.com.cn**

ANTENNA TESTING · INTERNATIONAL BUSINESS

Dongguan Voyager Telecom Equipment CO., LTD.

Room 1607, Building 1, No. 15 Gongye West Road
Songshan Lake Park, Dongguan, Guangdong, China

HEADQUARTERS

Xi'an Voyager Telecom Equipment CO., LTD. (Joint Stock)

6F, Voyager Building, No. 15 Gaoxin 2nd Road
High-tech Zone, Xi'an, Shaanxi, China
Tel: +86 29 8822 0419

ANTestify is Voyager's brand for base station antenna test systems.